

EARTHQUAKE INDICATOR

Using Arduino

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An earthquake is an unavoidable and unpredictable natural phenomenon that often causes damage to lives and property. We cannot fight it but we can stay alert and aware using

technology that can protect us and the industry. Here a simple earthquake indicator for

home and industry using an Arduino and a highly-sensitive ADXL335 (Fig. 1) accelerometer is presented that can indicate vibrations.

This project can be modified and used as a knock-and-shake detector for ATMs, vehicles or door-break alarms. But its main aim is to detect earthquakes and other seismic activities.

We know that accelerometers like ADXL335 are highly sensitive to knocks and vibrations in any of the three physical axes. ADXL335 gives analogue voltage equivalent

to imposed acceleration. It has three outputs, one each for of X-, Y- and Z-axes. The three analogue outputs are wired to Arduino Uno ADC pins. Any acceleration caused due to movement in any of the axes is detected by the accelerometer and hence by Arduino ADC.

If motion is violent enough during an earthquake and crosses a certain threshold, a local alarm light (LED) glows, a buzzer sounds as well as a relay energises. While the buzzer and light is for home purpose, relay output is for industrial

purpose; it can be wired to a PLC for safety interlocking of any moving machine part and furnace control for shutting these down in case of an earthquake. The threshold adjustment buttons are there for carrying out this task. An LCD has been provided for viewing threshold adjustments and for making the system user-friendly.

Circuit and working

The circuit (Fig. 2) uses Arduino Uno board wired to ADXL335 accelerometer module (connected across CON2) with its ADC inputs, namely, X-axis to A0, Y-axis to A1 and Z-axis to A2. Two pushbuttons through supply of 5V are wired to Arduino Uno interrupt pins 2 and 3 that are pulled down to ground via resistors R2 and R1. These buttons are used for incrementing and decrementing the threshold of vibration detection. A 16x2



Fig. 1: ADXL335

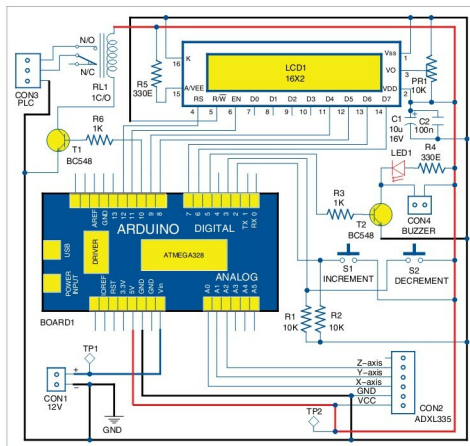


Fig. 2: Circuit diagram of the earthquake indicator